Binomial Dist.

10/14

Counting the number of green lights I hit on my way to work out of 5.

Assumptions: 1) First light 50/50.

2) $G_2 \rightarrow G$ is 90%.

3) $R \rightarrow$ 50/50. R/G.

$P(GGGGR) = 0.5 \times 0.9 \times 0.9 \times 0.9 \times 0.1$

Real Assump: 1) each traffic light is independent.

2) all lights have the same prob of success. (getting a green light) is p.. $P(G) = p$.

$P(GGGGR) = p \times p \times p \times p \times (1-p)$

multiply because ind events.

$= p^4(1-p)$

GRGGG, RGGGG, GGRGG, GGGRG.

$P($ 4 green lights out of 5 $)$ (no. of success)

$= 5 \times p^4(1-p)$.

$5 =$ combination term.

$X =$ no. of green lights out 5.

$X \sim \text{Binom}(n, p)$ ← parameters

X follows

where, n = no. of trials.

p = Prob of success.

Binomial dist satisfies 4 main assumptions:

B : Binary outcomes in each trial.

I : Independent trials.

N : No of trials is fixed.

S : Same probability of success in each trial.

$$P(X=k) = {}^{n}C_{k} \times p^{k} \times (1-p)^{n-k}$$

4 success, 1 fail

$$P(X=4) = P\begin{pmatrix} SSSSF \\ \vdots \\ FSSSS \end{pmatrix} = 5 \times p \times p \times p \times p \times (1-p)$$

5 cy

$k=4$

↳ combination term.

$n C_k$ = n choose k (R func: choose(n,k))

$n! = n(n-1)(n-2)\cdots 1$

$$n C_k = \frac{n!}{(n-k)!\,k!}$$

$$5C_4 = \frac{5!}{(5-4)!\,4!} = \frac{5!}{1!\,4!}$$

$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{1 \times 4 \times 3 \times 2 \times 1}$$

$$= 5$$

$$P(X=k) = {}^{n}C_{k}\, p^{k}\, (1-p)^{n-k}$$

diff comb of k success.

prob of success

prob of failure

$n \rightarrow$ no.

k

X = no. of heads I get in 3 coin flips.

actual outcome $\Rightarrow$ { HHH, HTH, HHT, THH, TTH, THT, HTT, TTT }

each (trial has H or T)

B : binary outcomes

I : independent

N : no. of trial is fixed ($n = 3$)

S : same prob of success in each trial ($p = 0.5$)

count of outcomes is success.

$S_X = \{3, 2, 1, 0\}$ ←

$$P(X = 2) = {}^{3}C_{2}\, p^{2} (1-p)^{1}$$

$$= 3(0.5)^{3}$$